Prayas JEE AIR (2025)

WEEKLY TEST 34

DURATION : 180 Minutes | **DATE : 20/01/2025** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (1)
3. (1)
4. (4)
5. (1)
6. (2)
7. (3)
8. (1)
9. (4)
10. (2)
11. (1)
12. (4)
13. (3)
14. (2)
15. (3)
16. (3)
17. (2)
18. (3)
19. (3)
20. (1)
21. (3)
22. (6)
23. (20)
24. (2)
25. (3)

CHEMISTRY

26. (3)
27. (2)
28. (2)
29. (1)
30. (2)
31. (2)
32. (2)
33. (3)
34. (3)
35. (1)
36. (2)
37. (1)
38. (4)
39. (3)
40. (3)
41. (2)
42. (4)
43. (3)
44. (2)
45. (2)
46. (3)
47. (128)
48. (25)
49. (2)
50. (4)

MATHEMATICS

51. (2)
52. (1)
53. (3)
54. (4)
55. (2)
56. (2)
57. (4)
58. (4)
59. (3)
60. (2)
61. (3)
62. (4)
63. (4)
64. (1)
65. (1)
66. (4)
67. (2)
68. (1)
69. (4)
70. (1)
71. (10)
72. (401)
73. (192)
74. (18)
75. (1)

SECTION-I (PHYSICS)

1. **(2)**

At maximum height, velocity is zero so,

$0 = u^2 - 2gh \;\; or \;\; u = \sqrt{2gh}$

As $s = ut + \frac{1}{2}at^2$ so, $\frac{h}{3} = \sqrt{2ght} - \frac{1}{2}gt^2$

$gt^2 - 2\sqrt{2ght} + \frac{2h}{3} = 0$

or $t = \frac{2\sqrt{2gh} \pm \sqrt{8gh - \frac{8gh}{3}}}{2g}$

$\frac{t_1}{t_2} = \frac{2\sqrt{2gh} - \sqrt{8gh - \frac{8gh}{3}}}{2\sqrt{2gh} + \sqrt{8gh - \frac{8gh}{3}}} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

2. **(1)**

Let additional force is $\vec{F}$, so that $F_{net} = 0$

$\vec{F}' + 5(+\hat{i}) + 6(-\hat{i}) + 7(+\hat{j}) + 8(-\hat{j}) = 0$

$\Rightarrow \;\; \vec{F}' - (\hat{i} + \hat{j}) = 0 \Rightarrow \vec{F} = \hat{i} + \hat{j}$

$\Rightarrow \;\; |\vec{F}| = \sqrt{1^2 + 1^2} = \sqrt{2}N$

$\theta = \tan^{-1}\left(\frac{\text{Component of } \hat{j}}{\text{Component of } \hat{i}}\right)$

$\theta = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ$

3. **(1)**

We know that $P = mV = \sqrt{2mKE}$

$\therefore \;\; 0.2u = \sqrt{2 \times 0.2 \times 90}$

$\Rightarrow \;\; u = 30\,\text{m/s}$

and $0.2v = \sqrt{2 \times 0.2 \times 40}$

$\Rightarrow \;\; v = 20\,\text{m/s}$

So, $a = \frac{v - u}{t} = \frac{20 - 30}{1} = \frac{-10\,\text{ms}^{-2}}{s^2}$

Now, using $x^2 - u^2 = 2aS$

$\Rightarrow \;\; S = \frac{0 - 30^2}{2 \times (-10)} = 45\,\text{m}$

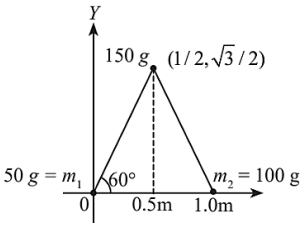
4. **(4)**

$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{m_1 + m_2 + m_3}$

$\vec{r}_{cm} = \frac{0 + 150 \times \left(\frac{1}{2}i + \frac{\sqrt{3}}{2}\hat{j}\right) + 100 \times \hat{i}}{300}$

$\vec{r}_{cm} = \frac{7}{12}\hat{i} + \frac{\sqrt{3}}{4}\hat{j}$

$\therefore$ Co-ordinate of center $= \left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)m$



5. **(1)**

Moment of inertia of solid sphere about any tangent,

$I = I_{cm} + mr^2 \; \frac{2}{5}mr^2 + mr^2$

$= \frac{7}{5}mr^2$

Moment of inertia of hollow sphere about any tangent,

$I = I_{cm} + mr^2$

$= \frac{2}{3}mr^2 + mr^2 = \frac{5}{3}mr^2$

M.O.I of circular ring about its diameter, $I = \frac{MR^2}{2}$

M.O.I of circular disc about any diameter, $I = \frac{MR^2}{4}$

6. **(2)**

Let d be the distance between the stars and let d_1 and d_2 be the distance of star from centre of mass Therefore,

$d_1 = \left(\frac{d}{m_1 + m_2}\right)m_2 \text{ and } d_2 = \left(\frac{d}{m_1 + m_2}\right)m_1$

Since, we have $\frac{Gm_1m_2}{d^2} = m_1\omega^2 d_1 = m_2\omega^2 d_2$

Also, $m_1 + m_2 = M$ {given}

$\Rightarrow d_1 = \frac{d}{M}m_2 \text{ and } d_2 = \frac{d}{M}m_1$

$\Rightarrow \frac{Gm_1(M - m_1)}{d^2} = m_1\omega^2\left[\frac{d}{M}(M - m_1)\right]$

$\Rightarrow GM = \omega^2 d^3$

$\Rightarrow d^3 = \frac{GMT^2}{4\pi^2}$

$\Rightarrow d = \left(\frac{GMT^2}{4\pi^2}\right)^{\frac{1}{3}}$

7. **(3)**

A. Torque $[\tau]=[F][r_\perp]$

$[\tau]=\left[MLT^{-2}\right][L]\Rightarrow\left[ML^2T^{-2}\right]$

B. $[\text{Stress}]=\dfrac{[F]}{[A]}=\dfrac{MLT^{-2}}{L^2}=\left[ML^{-1}T^{-2}\right]$

C. $[\text{Pressure gradient}]=\dfrac{[\Delta P]}{[\Delta X]}=\dfrac{\left[ML^{-1}T^{-2}\right]}{L}$

$\Rightarrow\left[ML^{-2}T^{-2}\right]$

D. Coefficient of viscosity $\Rightarrow F=6\pi\eta rv$

$[\eta]=\dfrac{[F]}{[r][v]}=\dfrac{\left[MLT^{-2}\right]}{[L]\left[LT^{-1}\right]}$

$[\eta]=ML^{-1}T^{-1}$

8. **(1)**

$\text{Stress}=\dfrac{\text{Weight}}{\text{Area}}$

$\Rightarrow\dfrac{10}{2.5\times10^{-4}}=\dfrac{25}{A}$

$\Rightarrow A=6.25\times10^{-4}$

9. **(4)**

As a neutron has more rest mass than a proton, so proton will require energy to decay into neutron.

10. **(2)**

W = Area under the cyclic curve

$W=\dfrac{1}{2}\times2m^3\times300\,Pa$

$W=300\,Pa-m^3$

W = 300 J

11. **(1)**

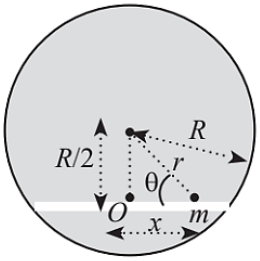
Force, $F = m\,E\cos\theta$

$=m\left(\dfrac{GM}{R^3}\times r\right)\times\dfrac{x}{r}$

$=\dfrac{mgR^2}{R^3}\times x$

$ma=\dfrac{mg}{R}x\Rightarrow a=\dfrac{g}{R}x$

Time period of SHM, $T=2\pi\sqrt{\dfrac{R}{g}}$



O ⇒ Mean Position

12. **(4)**

$y=\overline{A+B}\cdot B$

13. **(3)**

$\because E_n=-\dfrac{13.6Z^2}{n^2}eV$

$\Rightarrow\Delta E=13.6Z^2\left[\dfrac{1}{2^2}-\dfrac{1}{4^2}\right]eV$

$=13.6\times(4)2\left(\dfrac{1}{4}-\dfrac{1}{16}\right)eV$

$\Rightarrow\Delta E=40.8eV$

14. **(2)**

Let, $Q_0 = C_0V_0$ = initial charge on capacitor

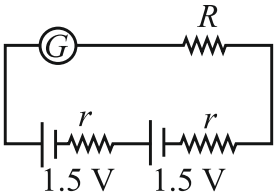
Battery connected means V remains unchanged

As separation id doubled, so $C=\dfrac{C_0}{2}$

$\Rightarrow U_1=\dfrac{1}{2}\left(\dfrac{C_0}{2}\right)V_0^2=\dfrac{C_0V_0^2}{4}$

$\Rightarrow\dfrac{U_1}{U_2}=\dfrac{1}{4}$

15. **(3)**

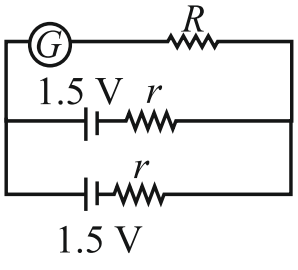


$I_1=1\text{A}$

$I_1R=3-2I_1r$

$I_1R+2I_1r=3$

$I_1=\dfrac{3}{R+2r}$ ....(i)



$I_2R=1.5-\dfrac{I_2}{2}r$

$I_2=\dfrac{1.5}{\left(R+\dfrac{r}{2}\right)}$

$I_2\left(R+\dfrac{r}{2}\right)=1.5$ ...(ii)

$I_2=0.6A$

solve (i) and (ii) r = 0.33 Ω

16. (3)

$$B_C = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2\left(x^2 + r^2\right)^{3/2}}$$

At $x = \frac{r}{2}$

$$B_a = \frac{\mu_0 I r^2}{2\left(\frac{r^2}{4} + r^2\right)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2\left(\frac{5}{4} r^2\right)^{3/2}} = \frac{\mu_0 I}{2r}\left(\frac{4}{5}\right)^{3/2}$$

$$= \frac{\mu_0 I}{2r}\left(\frac{2}{\sqrt{5}}\right)^3 = \left(\frac{2}{\sqrt{5}}\right)^3 B$$

17. (2)

$$\frac{1}{2} L i^2 = \frac{1}{n}\frac{1}{2} L i_0^2$$

$$\Rightarrow \quad i = \frac{i_0}{\sqrt{n}}$$

During growth of current,

$$i = i_0\left(1 - e^{-t/\tau}\right)\left(\tau = \frac{L}{R}\right)$$

$$e^{t/\tau} = \frac{\sqrt{n}}{\sqrt{n} - 1}$$

$$t = \frac{L}{R} \ln\left(\frac{\sqrt{n}}{\sqrt{n} - 1}\right)$$

18. (3)

The impedance, $Z = \frac{V_{rms}}{i} = \frac{200}{2} = 100\Omega$

We know that Power, $P = \frac{V_{\text{rms}}^2 R}{Z^2}$

or $320 = \frac{200^2 \times R}{100^2}$

$\therefore \quad R = 80\Omega$

We know that $Z = \sqrt{R^2 + X_L^2}$

Or $100 = \sqrt{80^2 + X_L^2}$

$\therefore \quad X_L = 60\Omega$

The back emf, $V_L = iX_L = 2 \times 60 = 120\,\text{V}$

19. (3)

EM wave is in direction $\rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Electric field is in direction $\rightarrow \hat{k}$

$E' \times B' \rightarrow$ direction of propagation of EM wave.

20. (1)

$$= \left(\frac{1}{3/2} - 1\right)\left(\frac{2}{R}\right)$$

$$= -\frac{2}{3R}$$

If a material of refractive index μ is filled in the cavity

$$= \left(\frac{\mu}{3/2} - 1\right)\left(\frac{2}{R}\right)\frac{2}{3R}$$

$$= \left(\frac{2\mu}{3} - 1\right)\frac{2}{R}1$$

$\Rightarrow 1 = 2\mu - 3\mu$

$\Rightarrow \mu = 2$

21. (3)

Formula of fringe width $(\beta) = \frac{D\lambda}{d} \Rightarrow \beta \propto \lambda$

$$\therefore \Rightarrow \frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \Rightarrow \beta_2 = \frac{\lambda_2}{\lambda_1} \times \beta_1$$

$$\Rightarrow \beta_2 = \frac{600\,\text{nm}}{400\,\text{nm}} \times 2\,\text{mm}$$

$$\Rightarrow \beta_2 = 3\,\text{mm}$$

22. (6)

Charged particle can be considered at the centre of a cube of side a, and the given surface represents its one side.

So, flux through each face $\phi = \frac{Q}{6\varepsilon_0}$

23. (20)

Volume capacity of beaker, V_0 = 500 cc

$V_b = V_0 + V_0 \gamma_{\text{beaker}} \Delta T$

When beaker is partially filled with V_m volume of mercury,

$V_m^1 = V_m + V_m \gamma_m \Delta T$

Unfilled volume $\left(V_0 - V_m\right) = \left(V_b - V_m^1\right)$

$$\Rightarrow V_0 \gamma_{\text{beaker}} = V_m \gamma_M$$

$$\therefore V_m = \frac{V_0 \gamma_{\text{beaker}}}{\gamma_M}$$

$$\text{or}, V_m = \frac{500 \times 6 \times 10^{-6}}{1.5 \times 10^{-4}} = 20\text{cc}.$$

24. (2)

Let ϕ_0 be theJ work function of the surface of the material. Then,

According to Einstein's Jphotoelectric equation, the maximum kinetic energy of the emitted

photoelectrons in the first case is $K_{max_1} = \frac{hc}{\lambda} - \phi_0$

and that in the second is

$$K_{max_2} = \frac{hc}{\frac{\lambda}{2}} - \phi_0 = \frac{2hc}{\lambda} - \phi_0$$

But $K_{max_2} = 3K_{max_1}$ (Given)

$$\therefore \quad \frac{2hc}{\lambda} - \phi_0 = 3\left(\frac{hc}{\lambda} - \phi_0\right)$$

$$\frac{2hc}{\lambda} - \phi_0 = \frac{3hc}{\lambda} - 3\phi_0$$

$$\therefore 3\phi_0 - \phi_0 = \frac{3hc}{\lambda} - \frac{2hc}{\lambda}$$

$$2\phi_0 = \frac{hc}{\lambda} \text{ or } \phi_0 = \frac{hc}{2\lambda}$$

25. (3)

$$3 \times \frac{v}{4l_c} = 4 \times \frac{v}{2l_0} \text{ or } \frac{l_c}{l_0} = \frac{3v}{4} \times \frac{2}{4v} = \frac{3}{8}$$

SECTION-II (CHEMISTRY)

26. (3)

$\Delta T_f \propto$ im (Assume molarity as molality)

$Na_2SO_4 = 3 \times 0.1$, urea $= 1 \times 0.2$, $Ca_3(PO_4)_2 = 5 \times 0.05$

0.3 : 0.2 : 0.25

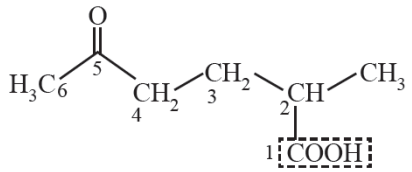
6 : 4 : 5

27. (2)

As in H_3PO_4 phosphorus is present in it's maximum oxidation state hence it cannot act as reducing agent.

28. (2)

In the given compound, the numbering will start from —COOH group. Double bond O present at 5th position will named as 'oxo' in prefix.



IUPAC name of the above compound is:

2–Methyl–5–oxohexanoic acid

29. (1)

$\left[Fe(H_2O)_6\right]Cl_2, Fe^{2+} \rightarrow 3d^6 \rightarrow (t_{2g})^4 (e_g)^2$

C.F.S.E. $= 4 \times (-0.4\Delta_o) + 2 \times 0.6\Delta_o = -0.4\Delta_o$

$K_2[NiCl_4], Ni^{2+} \rightarrow 3d^8 \rightarrow (e)^4 (t_2)^4$

C.F.S.E. $= 4 \times (-0.6\Delta_t) + 4 \times (0.4\Delta_t) = -0.8\Delta_t$

30. (2)

Higher the electron density on benzene ring, Higher the reactivity towards electrophileic substitution.

31. (2)

3° alcohols react fastest with $ZnCl_2$/conc.HCl due to formation of 3° carbocation and 2–methyl propan–2–ol is the only 3° alcohol.

32. (2)

$Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

$K_c = 10^{15} \times 10$

$E^o_{cell} = ?$

$$E^o_{cell} = \frac{2.303RT}{2F} \log\left(10^{15} \times 10\right)$$

$$= \frac{0.059}{2} \times [16 \log 10] \qquad [\log 10 = 1]$$

$E^\circ_{cell} = 0.4736\,V$

33. (3)

The empirical formula for the compound A can be calculated as:

	Mass	Moles $= \frac{\text{Weight}}{\text{Molar Mass}}$	Simplest Ratio
C	74g	$\frac{74}{12} = 6.17$	$\frac{6.17}{1.24} = 4.9 \approx 5$
H	8.7g	$\frac{8.7}{1} = 8.7$	$\frac{8.7}{1.24} = 7$
N	17.3g	$\frac{17.3}{14} = 1.24$	$\frac{1.24}{1.24} = 1$

Hence, empirical formula is C_5H_7N and empirical mass is 81 g/mol

34. (3)

On moving down a group, Electronegativity decreases and atomic radius increases for representative elements.

35. (1)

Rate law for the reaction, $-\frac{d[B_2]}{dt} = k[A][B_2]$

$$\therefore \ k = \frac{1.8 \times 10^{-3}}{0.015 \times 0.15} = 0.8$$

$$= -\frac{1}{2}\frac{d[A]}{dt} = -\frac{d[B_2]}{dt} = \frac{1}{2}\frac{d[AB]}{dt} = k[A][B_2]$$

$$= 0.8 \times 0.02 \times 0.04$$

$$\frac{d[AB]}{dt} = k[A][B_2] \times 2$$
$$= 0.8 \times 0.02 \times 0.04 \times 2$$
$$= 1.28 \times 10^{-3}\ M\ min^{-1}$$

36. (2)

I. $n + \ell = 4 + 1 = 5$

II. $n + \ell = 4 + 0 = 4$

III. $n + \ell = 3 + 2 = 5$

IV. $n + \ell = 3 + 1 = 4$

I is having greater n value as compared to III.
II is having greater n value as compared to IV.

37. (1)

$Ba(OH)_2\ (s) \rightleftharpoons Ba^{2+}\ (aq) + 2OH^-\ (aq)$

pH = 12 $\quad\quad\quad\quad$ $\therefore$ pOH = 2

i.e. $[OH^-] = 10^{-2}$

$$\left[Ba^{2+}\right] = \frac{10^{-2}}{2} = 5 \times 10^{-3}$$

$$K_{sp} = \left[Ba^{2+}\right]\left[OH^-\right]^2 = 5 \times 10^{-3} \times (10^{-2})^2$$

$= 5 \times 10^{-7}$

38. (4)

A & B $\rightarrow$ enantiomers
A & C $\rightarrow$ diastereomers

39. (3)

Guanidine is the most basic compound among the given compounds. Its conjugate acid is most stable due to resonance.

40. (3)

(a) Maltose is reducing sugar and gives silver mirror with tollen's reagent and has α (1, 4)-glyosidic linkage
(b) Sucrose, starch and cellulose are non-reducing sugar.
(c) Starch and cellulose are biopolymers.
(d) Sucrose has α,β (1,2)- glycosidic linkage.
(e) Starch and maltose have α-(1,4)-glycosidic linkage.
(f) Cellulose has β-(1,4)-glycosidic linkage.

41. (2)

Ease of decardoxylation depends upon stability of carbanion formed after decardoxylation

42. (4)

In isoelectronic species, lesser the negative charge more will be the ionisation energy, so Cl^- has more ionisation energy than S^{2-}.

43. (3)

Because Cr(+VI) has d° configuration hence d-d transition will not possible.

44. (2)

Atomic radius of Ga is slightly lower than that of Al due to d-Contraction.

45. (2)

$$CH_3CH_2CH_2Br \xrightarrow[80°C]{alc.\ NaOH}$$

$$CH_3 - CH = CH_2 \xrightarrow[AcOH]{HBr} CH_3 - \underset{\underset{Br}{|}}{CH} - CH_3$$

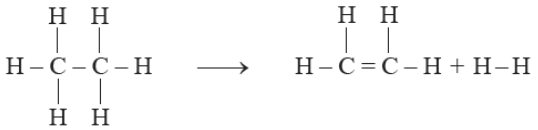
46. (3)

ii, iv, v $\rightarrow$ order = molecularity

47. (128)

$C_2H_6 \rightarrow C_2H_4 + H_2$

When bond is formed, Energy is released
When bond is broken, Energy is consumed



$$\Delta H_f = \left[\Delta H_{C-C} + 6\Delta H_{C-H}\right] - \Delta H_{C=C} - 4\Delta H_{C-H} - \Delta H_{H-H}$$
$$= \Delta H_{C-C} + 2\Delta H_{C-H} - \Delta H_{C=C} - \Delta H_{H-H}$$

Given: $\Delta H_{C-C} = 347\ kJ/mol$
$\Delta H_{C-C} = 611\ kJ/mol$
$\Delta H_{C-H} = 414\ kJ/mol$
$\Delta H_{H-H} = 436\ kJ/mol$
$= 347 + 2 \times 414 - 611 - 436$
$= 128\ kJ/mol$

48. (25)

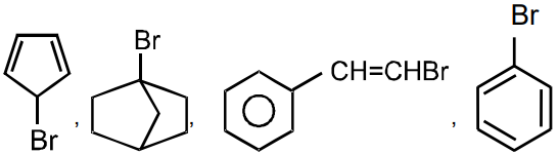
$MnO_4^- + 8H + 5e^- \longrightarrow Mn^{2+} + 4H_2O \quad E^\circ = 1.51V$

$\because$ 1 mole of MnO_4^- required 5 moles of electrons or 5F electricity
5 moles required = 25F

49. (2)

BrF_5, IF_5

50. (4)



are inert towards nucleophilic substitution by H_2O in aqueous solution.

SECTION-III (MATHEMATICS)

51. (2)

we have $64\cdot 9^x - 84\cdot 12^x + 27\cdot 16^x = 0$

$\Rightarrow 64\cdot\left(\frac{9}{16}\right)^x - 84\left(\frac{12}{16}\right)^x + 27 = 0$

$\Rightarrow 64\left(\frac{3}{4}\right)^{2x} - 84\left(\frac{3}{4}\right)^x + 27 = 0$

Let $\left(\frac{3}{4}\right)^x = t$

$\Rightarrow 64t^2 - 84t + 27 = 0$

$\Rightarrow (4t-3)(16t-9) = 0$

$\therefore t = \frac{3}{4}, \frac{9}{16}$

When $t = \frac{3}{4} \Rightarrow \left(\frac{3}{4}\right)^x = \frac{3}{4} \Rightarrow x = 1$

$t = \frac{9}{16} \Rightarrow \left(\frac{3}{4}\right)^x = \left(\frac{3}{4}\right)^2 \Rightarrow x = 2$

$\therefore \alpha = 1 \,\&\, \beta = 2$

Now $\alpha + \beta = 3$ and $\beta - \alpha = 1$

So, x^2 – (sum of roots) x + Product of roots = 0

$\Rightarrow x^2 - 4x + 3 = 0$

52. (1)

Let $I = \int \frac{3+2\cos x}{(2+3\cos x)^2}dx$, Multiplying N^r and D^r by $\text{cosec}^2 x$, we get

$\Rightarrow I = \int \frac{\left(3\text{cosec}^2 x + 2\cot x \text{cosec}\, x\right)}{\left(2\text{cosec}\, x + \cot x\right)^2}dx$

$= -\int \frac{-3\,\text{cosec}^2 x - 2\cot x \text{cosec}\, x}{\left(2\text{cosec}\, x + 3\cot x\right)^2}dx$

$= \frac{1}{2\text{cosec}\, x + 3\cot x} + C = \left(\frac{\sin x}{2+3\cos x}\right) + C$

53. (3)

$\log_2(x-2) \le \log_{1/2}(x+1)$

$x - 2 > 0, \qquad x + 1 > 0$

$x > 2 \qquad ...(1)$

$\log_2(x-2) \le \log_2 \frac{1}{x+1}$

$\Rightarrow \quad x - 2 \le \frac{1}{x+1} \quad \Rightarrow \quad \frac{x^2 - x - 3}{x+1} \le 0$

$\Rightarrow \quad x \in \left(-\infty, \frac{1-\sqrt{13}}{2}\right] \cup \left(-1, \frac{1+\sqrt{13}}{2}\right] \quad ...(2)$

by (1) and (2)

$\Rightarrow \quad x \in \left(2, \frac{1+\sqrt{13}}{2}\right]$

54. (4)

$\lim_{n\to\infty}\left[1 + \frac{\sum r(n-r+1)}{n^2(n+1)^2/4}\right]^n$

$= e^{\frac{n\left[n^2(n+1)/2 - n(n+1)(2n+1)/6 + n(n+1)/2\right]}{n^2(n+1)^2/4}}$

$= e^{\frac{1/2 - 1/3}{1/4}} = e^{\frac{2}{3}}$

55. (2)

$A = \{1, 2, 3\}$

$B = \{2, 3, 4\}$

$C = \{4, 5, 6\}$

$\therefore \quad B \cap C = \{4\}$

Hence, $A \times (B \cap C) = \{(1, 4), (2, 4), (3, 4)\}$

56. (2)

$2x^y + 3y^x = 20$

$2x^y\left(y\cdot\frac{1}{x} + \ln x \frac{dy}{dx}\right) + 3y^x\left(x\frac{1}{y}\cdot\frac{dy}{dx} + \ln y \cdot 1\right) = 0$

Put (2, 2)

$2.4\left(1 + \ln 2\frac{dy}{dx}\right) + 3.4\left(1.\frac{dy}{dx} + \ln 2\right) = 0$

$\frac{dy}{dx}\left[8\ln 2 + 12\right] + 8 + 12\ln 2 = 0$

$\frac{dy}{dx} = -\left[\frac{2+3\ln 2}{3+2\ln 2}\right] = -\left[\frac{2+\ln 8}{3+\ln 4}\right]$

57. (4)

$\because \quad x^n - y^n = (x-y)\,[x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \ldots. + y^{n-1}]$

$x^n - y^n$ is divisible by $x - y$

Stat 1 $\rightarrow (2023)^{2022} - (1999)^{2022}$

$(2023) - (1999) = 24$

$\therefore \quad (2023)^{2022} - (1999)^{2022}$ is divisible by 8

Start 2 $\rightarrow 13\,(1+12)^n - 11n - 13$

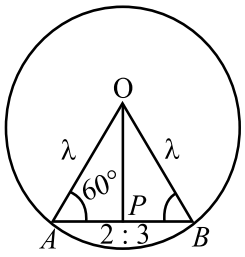
$13\,[1 + {}^nC_1\,(12) + {}^nC_2(12)^2 + \ldots..] - 11n - 13$

$\Rightarrow \quad (156\,n - 11n) + 13\cdot {}^nC_2\,(12)^2 + 13\cdot {}^nC_3\,(12)^3 + \ldots$

$\Rightarrow \quad (145n + 13\cdot {}^nC_2\,(12)^2 + 13\cdot {}^nC_3\,(12)^3 + \ldots)$

If $(n = 144m,\ m \in N)$ then it is divisible by 144 for infinite values of n.

58. (4)



Since OAB form equilateral Δ

$\therefore \angle OAP = 60°$

$AP = \frac{2\lambda}{5}$

$\cos 60° = \frac{OA^2 + AP^2 - OP^2}{2OA.AP}$

$\Rightarrow \quad \frac{1}{2} = \frac{\lambda^2 + \frac{4\lambda^2}{25} - OP^2}{2\lambda\left(\frac{2\lambda}{5}\right)}$

$\Rightarrow \quad \frac{2\lambda^2}{5} = \lambda^2 + \frac{4\lambda^2}{25} - OP^2$

$\Rightarrow \quad OP^2 = \frac{19\lambda^2}{25}$

$\Rightarrow \quad OP = \frac{\sqrt{19}}{5}\lambda$

Therefore, circle of radius is $\frac{\sqrt{19}}{5}\lambda$

59. (3)

$|\vec{a}| = \sqrt{11}, |\vec{c}| = \sqrt{22}$

$|\vec{a}| = |\vec{b} \times \vec{c}| = |\vec{b}||\vec{c}|\sin\theta$

$\sqrt{11} = \sqrt{50}\sqrt{22}\sin\theta$

$\Rightarrow \quad \sin\theta = \frac{1}{10}$

$|\vec{b} + \vec{c}|^2 = |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c}$

$= |\vec{b}|^2 + |\vec{c}|^2 + 2|\vec{b}||\vec{c}|\cos\theta$

$= 50 + 22 + 2 \times \sqrt{50} \times \sqrt{22} \times \frac{\sqrt{99}}{10}$

= 72 + 66

$\left|72 - |\vec{b} + \vec{c}|^2\right| = 66$

60. (2)

Here one women will sit between two men. Now fixing the place of one man the remaining 6 men on the circular table can sit in 6! ways. Since there are seven places between 7 men. Therefore seven women can sit on these places in 7! ways.

Thus 7 men and 7 women under the given condition can sit in 7! · 6! ways.

61. (3)

$a, ar, ar^2, \ldots.$

$T_2 + T_6 = \frac{25}{2} \Rightarrow ar\left(1 + r^4\right) = \frac{25}{2}$

$a^2r^2\left(1 + r^4\right)^2 = \frac{625}{4} \qquad \ldots\ldots(1)$

$T_3 \cdot T_5 = 25 \Rightarrow \left(ar^2\right)\left(ar^4\right) = 25$

$a^2r^6 = 25 \qquad \ldots\ldots(2)$

On dividing (1) by (2)

$\frac{\left(1 + r^4\right)^2}{r^4} = \frac{25}{4}$

$4r^8 - 17r^4 + 4 = 0$

$\left(4r^4 - 1\right)\left(r^4 - 4\right) = 0$

$r^4 = \frac{1}{4}, 4$

$\Rightarrow r^4 = 4$ (an increasing geometric series)

$a^2r^6 = 25 \Rightarrow \left(ar^3\right)^2 = 25$

$T_4 + T_6 + T_8 = ar^3 + ar^5 + ar^7$

$= ar^3 (1 + r^2 + r^4)$

$= 5(1 + 2 + 4) = 35$

62. (4)

$\frac{|x+1|}{|x|} + |x+1| = \frac{|x+1|^2}{|x|} \Rightarrow |x+1|$

$\left\{\frac{1}{|x|} + 1 - \frac{|x+1|}{|x|}\right\} = 0$

$\therefore |x+1| = 0$ or $1 + |x| - |x+1| = 0.$

$|x+1| = 0 \Rightarrow x = -1.$

If $x < -1, 1 + |x| - |x+1| = 0$

$\Rightarrow 1 - x + x + 1 = 0 \Rightarrow 2 = 0$

If $-1 \le x < 0, 1 + |x| - |x+1| = 0$ (absurd).

$\Rightarrow 1 - x - (x+1) = 0 \Rightarrow x = 0$ (not possible)

$x \ge 0, 1 + x - (x+1) = 0 \Rightarrow 0 = 0 \Rightarrow x$ can have any value in the interval.

63. (4)

$\tan A + \tan B + \tan C = k$

$\left[\frac{\tan A + \tan B}{1 - \tan A \tan B}\right] [1 - \tan A \tan B] + \tan C = k$

$\tan(A + B) [1 - \tan A \tan B] + \tan C = k$

$\tan(180° - C) [1 - \tan A \tan B] + \tan C = k$

$- \tan C [1 - \tan A \tan B] + \tan C = k$

$\therefore \tan A \tan B \tan C = k.$

64. (1)

Determinant of the coefficients

$= \begin{vmatrix} \alpha & -3 & 1 \\ 1 & \alpha & 3 \\ 3 & 1 & 5 \end{vmatrix} = 5\alpha^2 - 6\alpha - 11 = (5\alpha - 11)(\alpha + 1)$

The system does not have a unique solution iff $\alpha = \frac{11}{5}$ or -1.

65. **(1)**

If $x = a$, where 'a' is an integer then

$f(a) = 2a + a + \frac{1}{2}\sin 2a.$

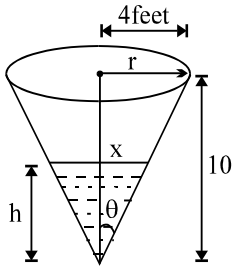
But $\lim_{h\to 0} f(a-h) = 2a + (a-1) + \frac{1}{2}\sin 2a$

$\therefore$ Values between $\lim_{h\to 0} f(a-h)$ and $f(a)$ are never achieved.

Also, $f'(x) = 2 + \cos 2x > 0$, i.e., $f(x)$ is strictly increasing.

66. **(4)**

$\frac{dV}{dt}$ = 2 cubic feet per minute; $\frac{dh}{dt}$ = ?



$V = \frac{1}{3}\pi x^2 h$

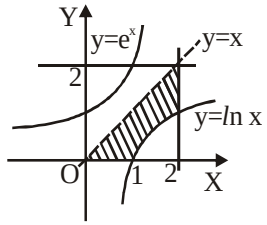
$\tan\theta = \frac{x}{h} = \frac{r}{10} = \frac{4}{10} \quad \Rightarrow \quad x = \frac{2h}{5}$

$V = \frac{1}{3}\cdot\pi\cdot\frac{4h^2}{25}\cdot h = \frac{4\pi}{75}\cdot h^3$

$\frac{dV}{dt} = 2 = \frac{4\pi}{75}3h^2\frac{dh}{dt}$; $\frac{dh}{dt} = \frac{150}{12\pi h^2} = \frac{150}{12\cdot\pi\cdot 25}$

$= \frac{6}{12\pi} = \frac{1}{2\pi}$

67. **(2)**



$A = 2\left(\frac{1}{2}\cdot 2\cdot 2 - \int_1^2 \ln\int x\,dx\right) = 6 - 2\ln 4.$

68. **(1)**

We have S = {1, 2, 3,.... , 100}

Let A : integer 2 is chosen.

B : even integer is chosen .

Given P(1) = λ ln 1 ; P(2) = λ ln 2 ; P(3) = λ ln 3 etc. (where λ is proportionality constant)

Now $P(A/B) = \frac{P(A\cap B)}{P(B)} = \frac{\lambda\ln 2}{\lambda[\ln(2.4.6....100)]}$

$= \frac{\ln 2}{[\ln(2^{50}\cdot 50!)]} = \frac{\ln 2}{50\ln 2 + \ln(50!)}$

(By multiplication rule)

69. **(4)**

For $R_1 = \phi$ so transtitive

For R_2, if $(x, y) = (0,1)$ and $(1,2)$

So false

For R_3, if $(x, y) = (0,1), (1,2)$ so true

For R_4, if $(x, y) = (1,0), (0,1)$ so true

70. **(1)**

Given differential equation

$\frac{dy}{dx} + (8 + 4\cot 2x)y = \frac{2e^{-4x}}{\sin^2 2x}(2\sin 2x + \cos 2x)$

I.F. = $\int(8 + 4\cot 2x)dy = e^{8x + 2\log_e(\sin 2x)}$

$= e^{8x}\cdot\sin^2 2x$

$\therefore$ Solution is

$y\left(e^{8x}\cdot\sin^2 2x\right) = \int 2e^{4x}(2\sin 2x + \cos 2x)dx + C$

$\therefore \quad y(x) = \frac{e^{-4x}}{\sin 2x}$

$\therefore \quad y\left(\frac{\pi}{6}\right) = \frac{e^{-4\frac{\pi}{6}}}{\sin\left(2\cdot\frac{\pi}{6}\right)} = \frac{2}{\sqrt{3}}e^{-\frac{2\pi}{3}}$

71. **(10)**

$f(x) = \frac{\sec^2 x + 4\tan x}{\sec^2 x}$

$= \frac{\tan^2 x + 1 + 4\tan x + 8}{1 + \tan^2 x}$

$= \frac{\tan^2 x + 4\tan x + 9}{1 + \tan^2 x}$

$= \frac{2\cdot(2\tan x)}{1 + \tan^2 x} + 4\frac{\left(1 - \tan^2 x\right)}{1 + \tan^2 x} + 5$

$= 2\sin 2x + 4\cos 2x + 5$

$R_f = [5 - \sqrt{20}, 5 + \sqrt{20}]$

72. **(401)**

$Q = PAP^T$

$P^T.Q^{2007}.P = P^T.Q.Q.....Q.P$

$= P^T(PAP^T)(P.AP^T).....(PAP^T)P.$

$\Rightarrow \quad (P^TP)A(P^TP)A.....A(P^TP)$

$$P^T.P=\begin{bmatrix}\sqrt{3}/2 & -1/2\\ 1/2 & \sqrt{3}/2\end{bmatrix}\begin{bmatrix}\sqrt{3}/2 & 1/2\\ -1/2 & \sqrt{3}/2\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=1$$

$\therefore \quad P^T.Q^{2007}.P=A^{2007}$

$$A^2=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$$

$$\therefore \quad A^{2007}=\begin{bmatrix}1 & 2007\\ 0 & 1\end{bmatrix}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

$a = 1, b = 2007, c = 0, d = 1$

$2a + b – 3c – 4d = 2 + 2007 – 4 = 2005$

73. (192)

$$e=\sqrt{1-\frac{3}{4}}=\frac{1}{2}$$

$$b^2=64(4-1)=192$$

74. (18)

If the lines $\frac{x-1}{2}=\frac{2-y}{-3}=\frac{z-3}{\alpha}$ and

$\frac{x-4}{5}=\frac{y-1}{2}=\frac{z}{\beta}$

Intersect Point of first line (1, 2, 3) and point on second line (4, 1, 0).

Vector joining both points is $-3\hat{i}+\hat{j}+3k$

Now vector along second line is $2\hat{i}+3\hat{j}+\alpha k$

Also vector along second line is $5\hat{i}+2\hat{j}+\beta k$

Now these three vectors must be coplanar

$$\Rightarrow \quad \begin{vmatrix}2 & 3 & \alpha\\ 5 & 2 & \beta\\ -3 & 1 & 3\end{vmatrix}$$

$\Rightarrow \quad 2(6 – \beta) – 3(15 + 3\beta) + \alpha(11) = 0$

$\Rightarrow \quad \alpha – \beta = 3$

Now $\alpha = 3 + \beta$

Given expression $8(3 + \beta).\beta = 8(\beta^2 + 3\beta)$

$$=8\left(\beta^2+3\beta+\frac{9}{4}-\frac{9}{4}\right)=8\left(\beta+\frac{3}{2}\right)^2-18$$

So magnitude of minimum value = 18

75. (1)

$\sin^{-1}(1 + x)$ is defined for $x < 0$ and

$$\sin^{-1}x=\frac{\pi}{2}-\cos^{-1}x \forall -1\le x\le 1$$

The given equation is $\sin^{-1}x + \sin^{-1}(1 + x) = \cos^{-1}x$

Which can be written as

$$\frac{\pi}{2}-\cos^{-1}x+\frac{\pi}{2}-\cos^{-1}(1+x)=\cos^{-1}x$$

$\Rightarrow \quad \pi – \cos^{-1}(1 + x) = 2\cos^{-1}x$

$\Rightarrow \quad \cos^{-1}(–1 – x) = 2\pi – \cos^{-1}(2x^2 – 1)$

$\Rightarrow \quad \cos^{-1}(–1 – x) + \cos^{-1}(2x^2 – 1) = 2\pi$

$\Rightarrow \quad \cos^{-1}(–1 – x) = \cos^{-1}(2x^2 – 1) = \pi$

$\Rightarrow \quad –1 – x = 2x^2 –1 = –1$

$\Rightarrow \quad x = 0$

Which implies that the total number of solutions $\sin^{-1}x + \sin^{-1}(1 + x) = \cos^{-1}x$ is only one.



PW Web/App - https://smart.link/7wwosivoicgd4

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